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RESEARCH PROFILE

As a resourceful and dedicated biochemist, I have over five years of experience in research using mass spectrometry. My passion lies in phytochemistry, metabolomics, and proteomics, and I have published my findings in highly renowned Q1 peer-reviewed journals. My expertise includes a strong background in a variety of chromatographic and spectrometric instruments, such as HPLC (Waters, Agilent, Shimadzu), UPLC (Waters), UPLC-MS/MS (Waters), UHPLC-DAD-QTOF-IMS (Agilent), MALDI-TOF-TOF-MS (Bruker), and GC-MS (Shimadzu). I also possess a solid understanding of various laboratory techniques, including spectrometric assays, separation techniques, ultra-sonication, and both qualitative and quantitative antioxidant assays. Additionally, I am proficient in scientific writing, publishing, and data analysis software like MetaboAnalyst, MS DIAL, XC-MS, MPP, SpectraMill, PAST, GRAPHPAD PRISM, and MeV, and I am skilled in both Linux and Windows operating systems.

RESEARCH INTERESTS

Metabolomics, proteomics and lipidomics using mass spectrometry with chemometric analysis.

Development of novel extraction, isolation and separation techniques for biomolecules.

Analytical Chemistry: Method development and validation.

Separation of isomers using Ion-Mobility Mass Spectrometry (IMS).

Nutri-metabolomics and its application.

Spatial biology using imaging mass spectrometry-DESI-MS, MALDI-MSI and FT-ICR.

EDUCATION

Doctor of Philosophy (Biological Sciences) Academy of Scientific & Innovative Research (AcSIR), CSIR-IHBT, Palampur, Himachal Pradesh, India (2020- 2025).

Master of Technology (Agricultural Biotechnology), Indian Institute of Technology-Kharagpur, Kharagpur, West Bengal, India (2018-2020).

Master of Science (Biotechnology), Central University of Haryana, Jant-Pali, Haryana, India (2016-2018).

Bachelor of Science (Pass course), Maharshi Dayanand University, Rohtak, India (2012-2016).

RESEARCH EXPERIENCE (2020-till date)

PhD (Biological Sciences) (Aug2020-Dec2025)
CSIR-IHBT, Palampur, HP, India
Method development and validation for aconites

1] Optimization of phytochemical analysis of *Aconitum heterophyllum* metabolites using UHPLC- IMS-QTOF.
2] Developed and validate a rapid, sensitive, and reproducible method for the simultaneous identification and quantification of eight alkaloids from *Aconitum heterophyllum* using UHPLC-DAD-QTOF-IMS.
3] Demonstrated multi-dimensional validation using Ion Mobility Spectrometry (IMS) for enhanced compound identification.

Therapeutic insights of *Aconitum heterophyllum* metabolites

1] Constructed a comprehensive metabolite reference map identifying 266 compounds across five tissues using UHPLC-MS, GC-MS, and MALDI-TOF techniques.
2] Performed molecular docking of identified metabolites with 42 disease-related proteins, revealing 80 bioactive compounds with potential roles in antioxidant, immunomodulatory, and other therapeutic pathways.
3] Validated the antioxidant and immunomodulatory effects of Tri-N-acetylchitotriose and Camellianin A through *in vitro* assays on RAW264.7 cells, demonstrating reduced ROS, restored mitochondrial membrane potential, and suppressed pro-inflammatory cytokines.

Light dependent aconite biosynthesis of *Aconitum heterophyllum* metabolites

1] Conducted a comprehensive transcriptomic and metabolomic study to uncover light-dependent regulation of aconite biosynthesis in *Aconitum heterophyllum*.
2] Identified 3,444 differentially expressed genes and key transcription factors (MYB, bHLH, NAC, WRKY) involved in alkaloid production.
3] Discovered and annotated a cytochrome P450 gene (*Ah*CytP450) as a potential regulator of diterpenoid alkaloid biosynthesis. Established a foundation for genetic interventions to enhance medicinal metabolite yield.

TECHNICAL COMPETENCIES

A. **Mass Spectrometry and Proteomics:** DIA/DDA, Medicinal Proteomics, Intact Protein Analysis, PTM Analysis.
B. **Analytical and Separation Chemistry:** High-Performance Liquid Chromatography (HPLC/UHPLC), LC MS/MS, Fast Protein Liquid Chromatography (AKTA pure™), Ultracentrifugation.
C. **Biochemistry and Cell Biology:** Mammalian Cell Culture, Enzyme Assays & Kinetics, Antimicrobial Screening.
D. **Computational Biology & Bioinformatics:** Molecular Docking (Discovery Studio, PyRx, AutoDock), Network biology and polypharmacology approaches (STITCH, STRING, Cytoscape), Mass Spectometeric Data Analysis (Progenesis QI, MS DIAL, XC-MS, MPP, SpectraMill), Statistical Analysis (MetaboAnalyst, PAST, GRAPHPAD PRISM, and MeV).
E. **Instrumentation:** UHPLC-QTOF-MS/MS (Agilent 6560 Ion Mobility), MALDI-TOF/TOF (Bruker), IDEAS ImageStream, QuantStudio 3, CFX Opus 96.

LIST OF PUBLICATIONS with impact factors 2024

1. Thakur, S., Kumari, S., **Punia, A**., Patial, V., Singh, D. and Kumar, R., 2024. Multi-omics analysis reveals the protective effect of a novel bioactive peptide (BP1) in cardiomyopathy using a zebrafish model. ***Journal of Proteome Research*** (IF-3.6)
2. **Punia, A.**, Satyakam., Saini, V., Chouhan, M., Chakraborty, N. and Kumar, R. 2026. Dancing to the tune of the light regime: from perception to secondary metabolite biosynthesis. ***Critical Reviews in Biotechnology*** **(IBTY),** 47, pp 1-21 (IF 7.7)
3. Thakur, S., **Punia, A**., Acharya, V., Kumar, B., Prasad, A., Yadav, S.K. and Kumar, R., 2024. Bringing bioactive peptides into drug discovery: Challenges and opportunities for medicinal plants. **Industrial Crops and Products**, 222, p.119855. (IF-6.2)
4. **Punia, A**., Kumari, M., Chouhan, M., Saini, V., Joshi, R., Kumar, A. and Kumar, R., 2024. Proteomic and metabolomic insights into seed germination of *Ferula assa-foetida*. **Journal of Proteomics**, 300, p.105176. (IF-2.8)
5. **Punia, A**., Joshi, R. and Kumar, R., 2022. Identification and quantification of eight alkaloids in *Aconitum heterophyllum* using UHPLC-DAD-QTOF-IMS: A valuable tool for quality control. **Phytochemical Analysis**, 33(7), pp.1121-1134. (IF-2.6)
6. Kumari, M., Pradhan, U.K., Joshi, R., **Punia, A**., Shankar, R. and Kumar, R., 2021. In-depth assembly of organ and development dissected *Picrorhiza kurroa* proteome map using mass spectrometry. **BMC Plant Biology**, 21, pp.1-18. (IF-4.8)

Book Chapters

1. Saini, V., **Punia, A.**, Choudhary, A., Vishal, Singh, P.K. and Jayaraman, N.K., 2024. Unlocking Ethnomedicinal Plant Potentials: Advancing Drug Discovery in the High-Throughput Omics Era. **In Ethnomedicinal Plants for Drug Discovery: Current Developments** (pp. 443-462). Singapore: Springer Nature Singapore.
2. **Punia, A.**, Satyakam and Kumar, R., 2025. Mass Spectrometry-Based Methods for Plant Metabolite Profiling. ***Plant Stress Memory: Methods and Protocols***, pp.103-117.

Scientific Publications Metrics

Total publications: **7**
Citation: **45** (Google Scholar)
h-index: **3** i-10 index: **3**
Book Chapter: **2**
Paper presented in conferences: **3**

CONFERENCES / POSTER PRESENTATION / ORAL PRESENTATIONS

- "**73rd Conference on Mass Spectrometry and Allied Topics**" organized by American Society for Mass Spectrometry at Baltimore, Maryland from June 1-5, 2025.
- "**8th STUDENT SEMINAR SERIES-2024**" held on the occasion of Teachers Day at CSIR-IHBT, Palampur on September 5th, 2024
- "**International Symposium on Plant Biology and Functional Genomics (ISPBFG 2024)**" held on March13rd-15th, 2024 conducted by University of Hyderabad, Hyderabad, Telangana India.
- "**BIOTECHNOLOGY: EXPLORING THROUGH INNOVATION**" held on November 13th - 14th, 2017 organized by Department of Biotechnology, Central University of Haryana, Jant-Pali, Haryana, India.
- "**BIOTECHNOLOGY FOR SUSTAINABLE DEVELOPMENT (BSD-2016)**" held on November 4th-5th, 2016 organized by Department of Biotechnology, Central University of Haryana, Jant-Pali, Haryana, India.

TRAININGS / SAMINARS / WORKSHOPS

- "**Mass Spectrometry Workshop**" held on April 3rd – 5th, 2026 organized by MetaChem Academy, India.
- "**DATA REPRODUCIBILITY WORKSHOP SERIES**" held on February 9th – 11th, 2026 organized by Resilience Institute, Michigan State University. United States.
- "**The 2025 Mass Spectrometry Digital Summit**" held on July 22nd and 23rd, 2025 organized by LabX Media Group, Midland, Ontario.
- "**Spatial Metabolomics Workshop**" held on March 5th, 2025 organized by Plant Resilience Institute, Michigan State University. United States.
- "**Research Ethics, Plagiarism Checking & Removal**" workshop held on February 17th, 2024 conducted by Research Graduate.
- "**The Broad Utility (and Potential Pitfalls) of qPCR in Diagnosing Infectious Diseases**" seminar held on October 25th, 2023 organized by Technology Networks, Sudbury, United Kingdom.
- "**Advances in Proteomics & Metabolomics**" seminar held on October 11th-12th, 2023 organized by Technology Networks, Sudbury, United Kingdom.
- "**IP Awareness/Training program**" held on April 22th, 2022 organized by Intellectual Property Office, India.

MEMBER OF PROFESSIONAL SOCIETIES

- **NORDIC Metabolomics Society**, Copenhagen, Denmark
- **Metabolomics Brazil Portal**, Brazil
- **Metabolomics South Africa**, South Africa

REFERENCES

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Biotechnology Division
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Dr. Robin Joshi is my PhD collaborator.